AE4353 Artificial Intelligence for Aerospace Control and Operations

Contrail detection using AI applied to Geostationary Satellite images

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This presentation



Motivation

Explain what contrails are, how they contribute to aviation's climate impact, and how we can mitigate this — 5 min.

Problem setup

Define the contrail detection task, and show how you can solve it for geostationary satellite images using deep learning — 5 min.

Historical context

Give an overview of recent developments in contrail detection, including the creation of the OpenContrails dataset (which you will use in your assignment) — 5 min.

Bonus

Explanation of the dice score and some tricks of the trade

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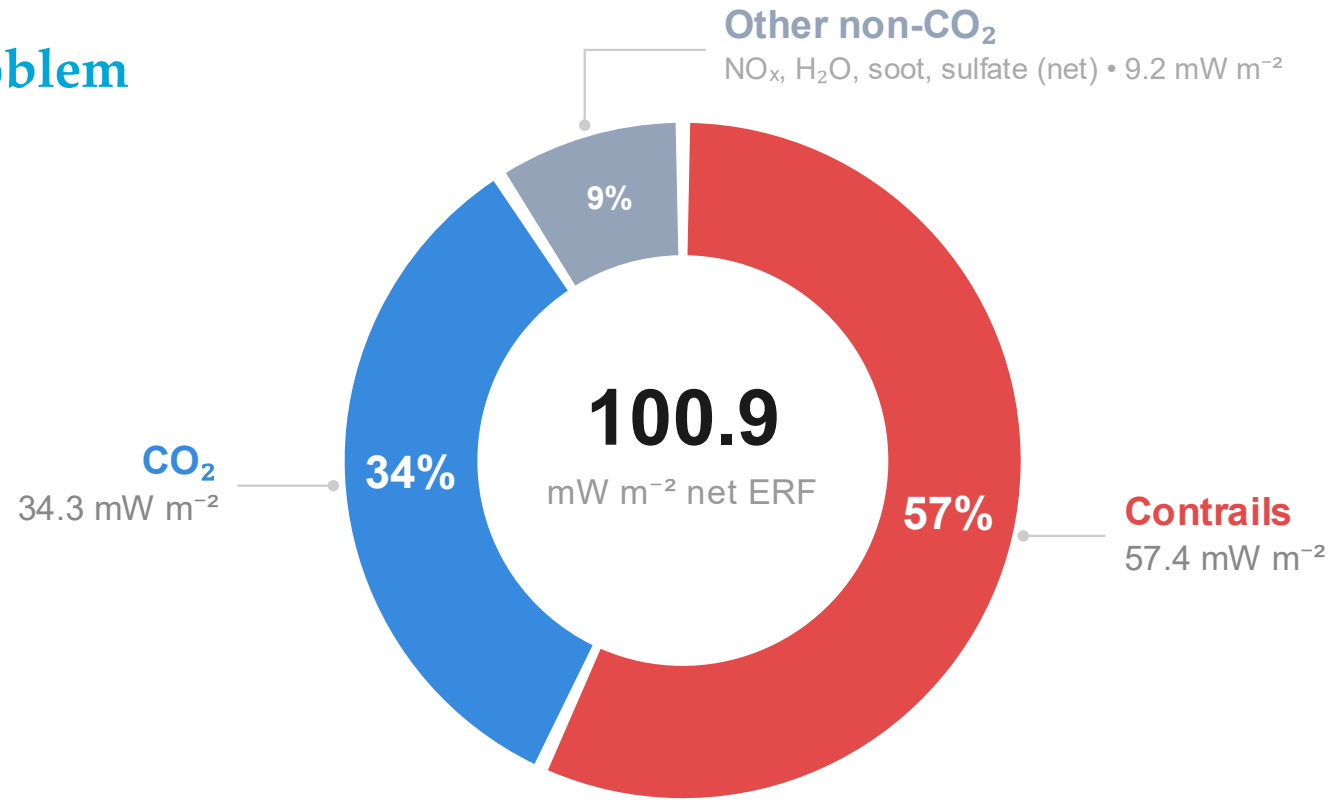
Motivation: most aviation climate "solutions" are decades away

3.5% of total net anthropogenic effective radiative forcing is estimated to be **caused by aviation**

Aviation is a **hard to de-carbonize sector:** technological solutions are **decades away**

CO_2 emissions **are literally only half of the problem**

If only there was a mitigation pathway for this **contrail impact** that we could implement within the next 5 years…



Lee et al. (2021), Atmos. Environ. 244, 117834

So what are these contrails?

Mixing of **warm, moist** engine exhaust with **cooler, drier** ambient air can lead to temporary **supersaturation of water** and thus **cloud formation**



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Which contrails matter?

Most contrails disappear quickly after they form...



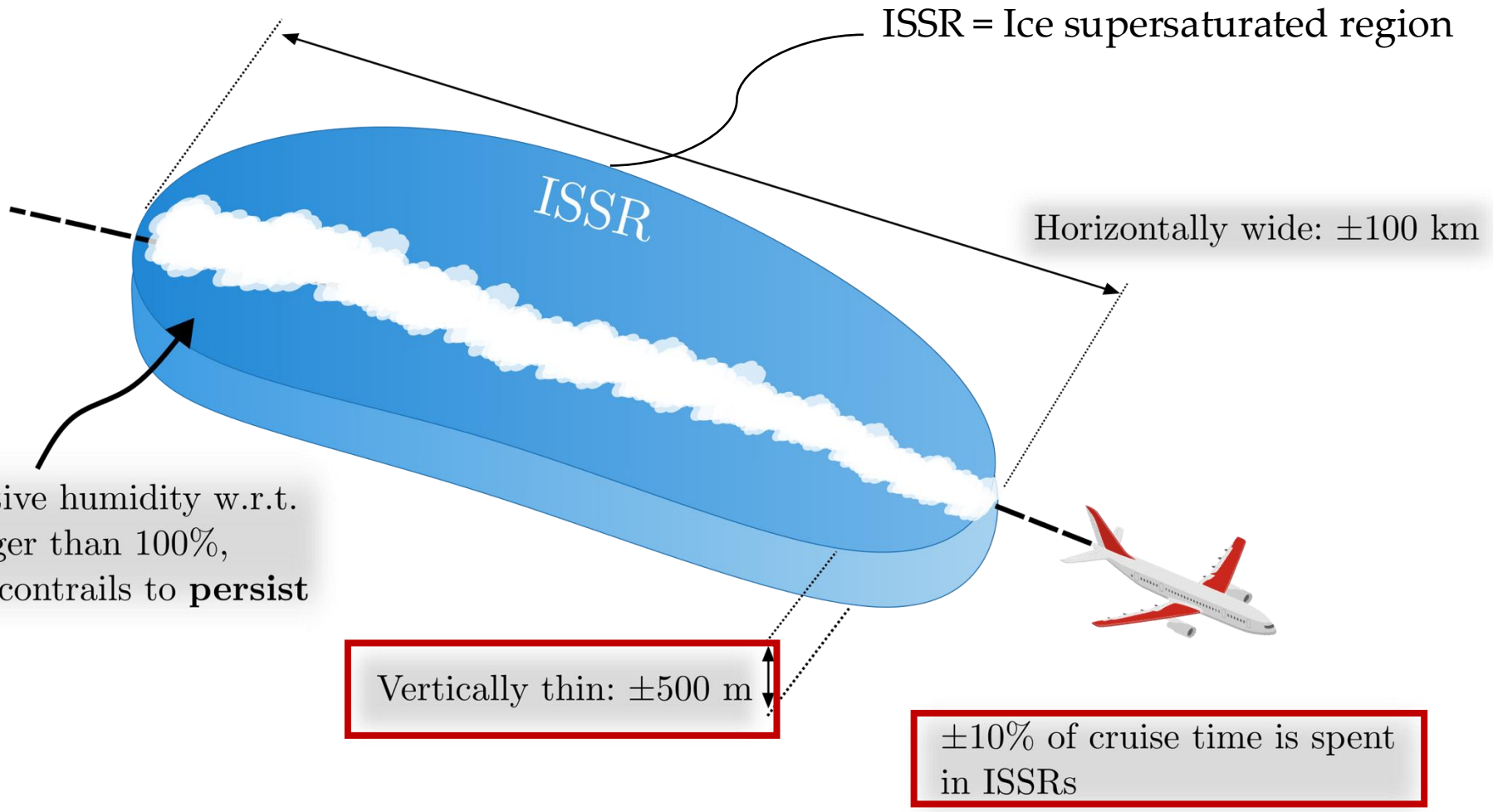
But if the air is sufficiently humid, they can persist for hours



These **persistent contrails** are the ones that warm the climate and they are the ones that we want to get rid of

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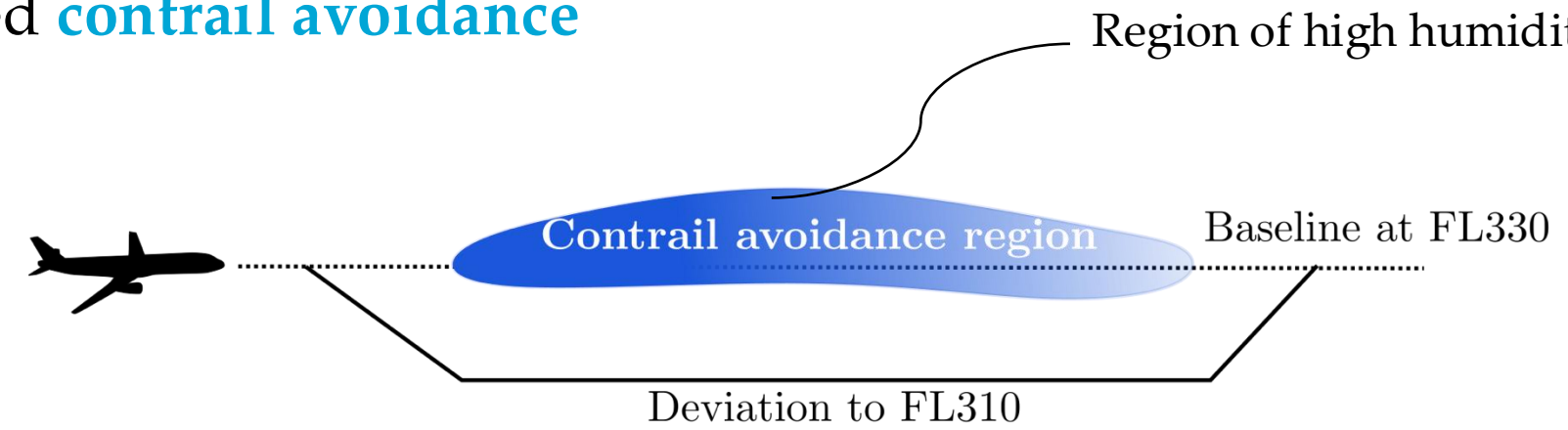
What do the regions where contrails persist look like?



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Can we avoid regions where contrails form and persist?

Yes: it's called **contrail avoidance**



This not just an idea: we have tested this with commercial aircraft and we know it works.

The limiting factor of contrail avoidance is our ability to predict where these high-humidity regions are: existing weather models are not good at this.

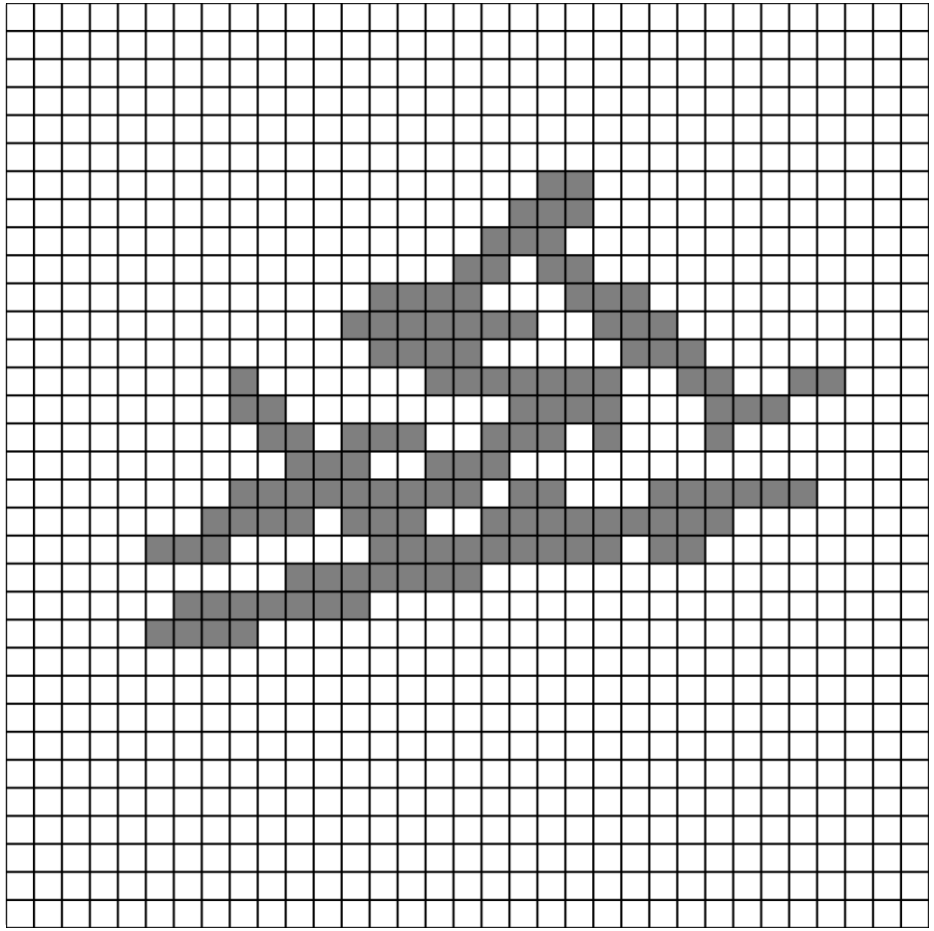
One solution: observe persistent contrails in geostationary satellite images and use this to improve our models and quantify the success of avoidance.

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What is contrail detection?

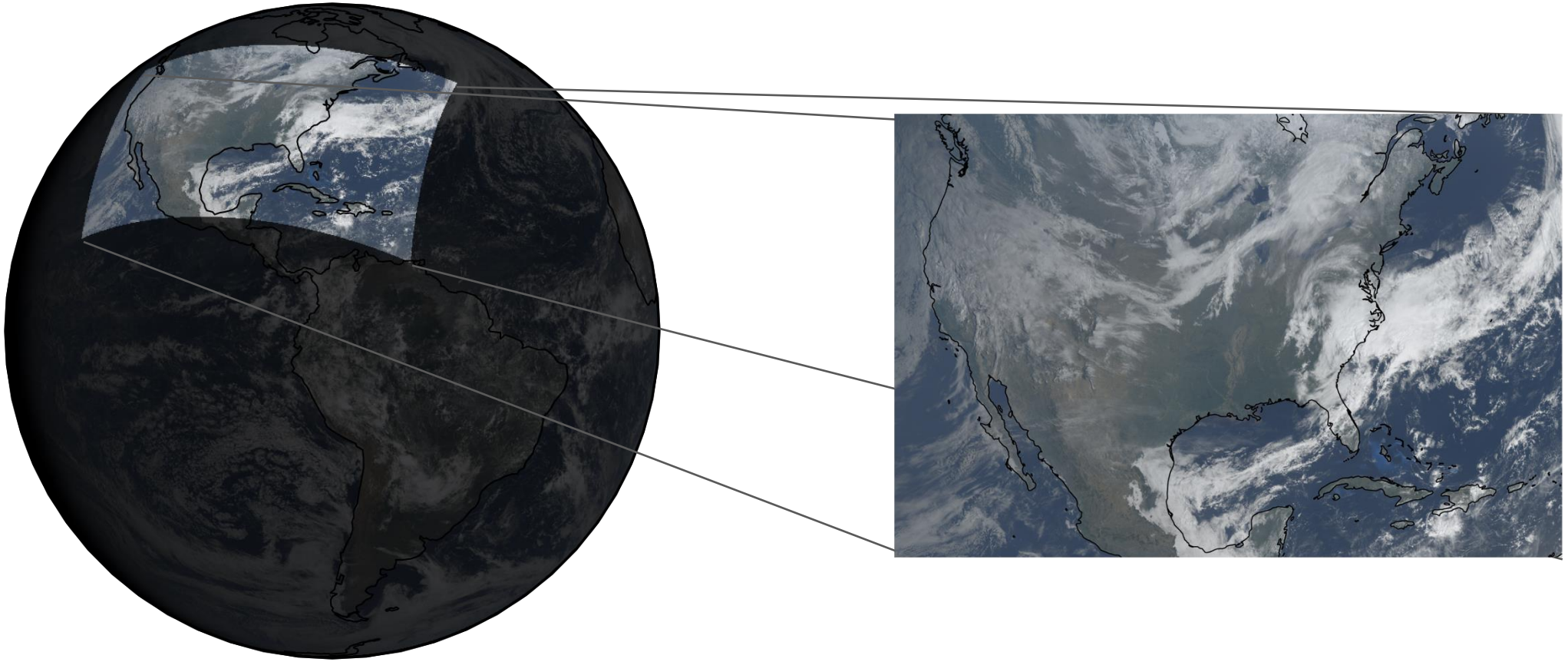
Contrail detection: determine which pixels in an image are part of a contrail and which are not.

In computer vision terms, contrail detection is **semantic segmentation** with two classes: contrail and not-contrail.



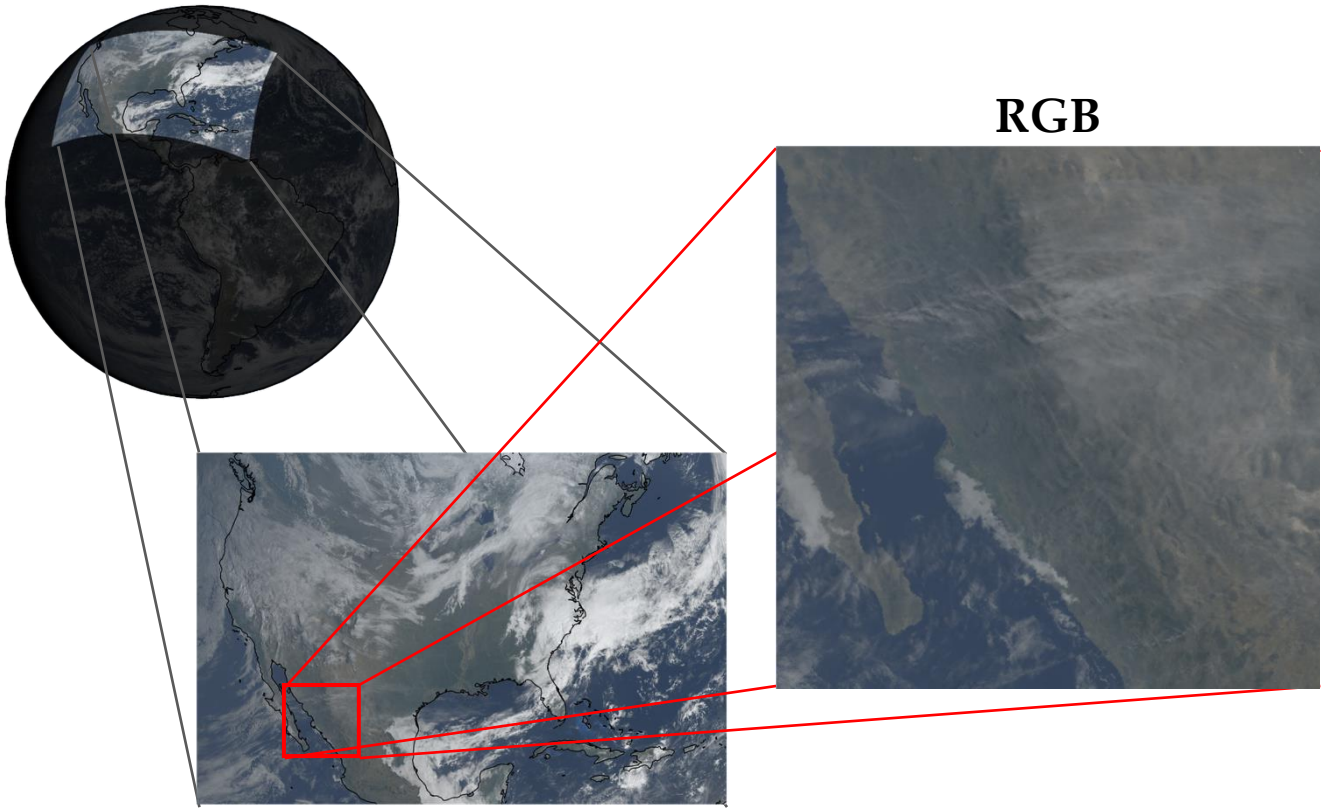
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Deep-learning based contrail detection in geostationary satellite imagery



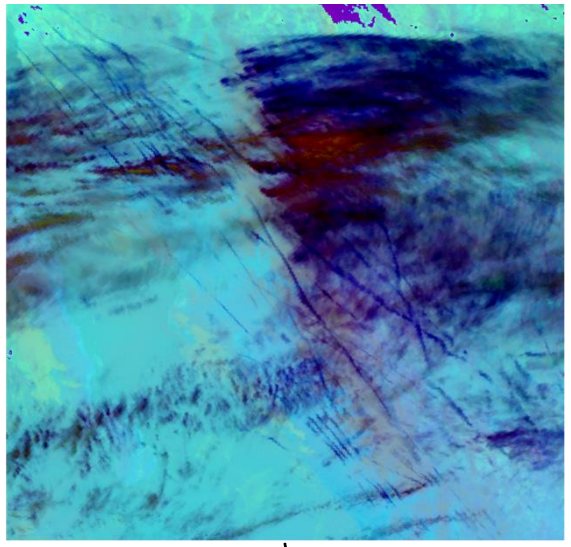
Meijer et al., 2022

Deep-learning based contrail detection in geostationary satellite imagery

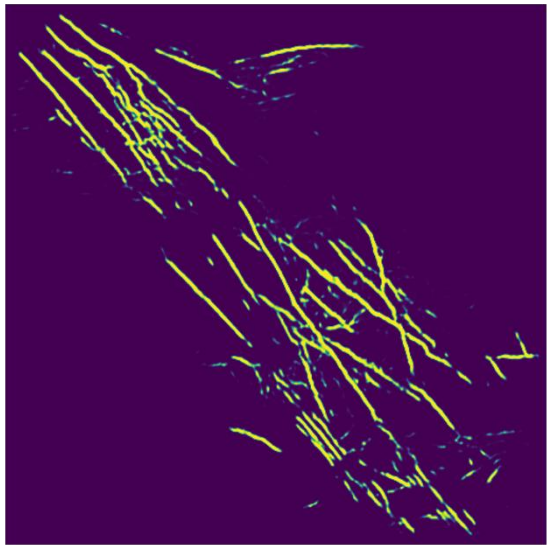


RGB

Infrared



Contrail detections

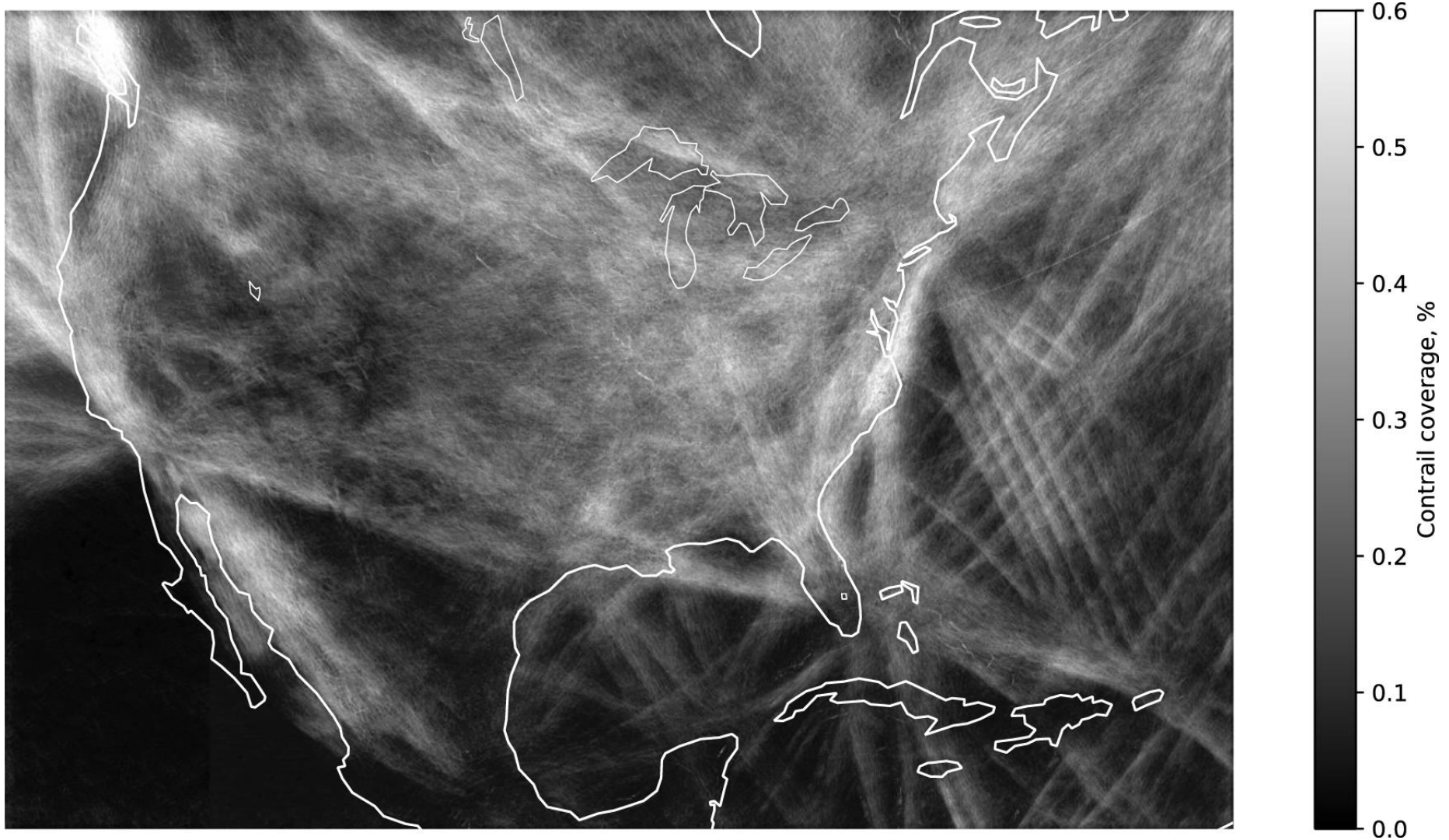


Convolutional Neural Network trained and evaluated with 100+ human labelled images

Meijer et al., 2022

Application: spatial distribution of contrail detections

Averaged over 2018 and 2019



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Figure 1 from Meijer et al. (2022)

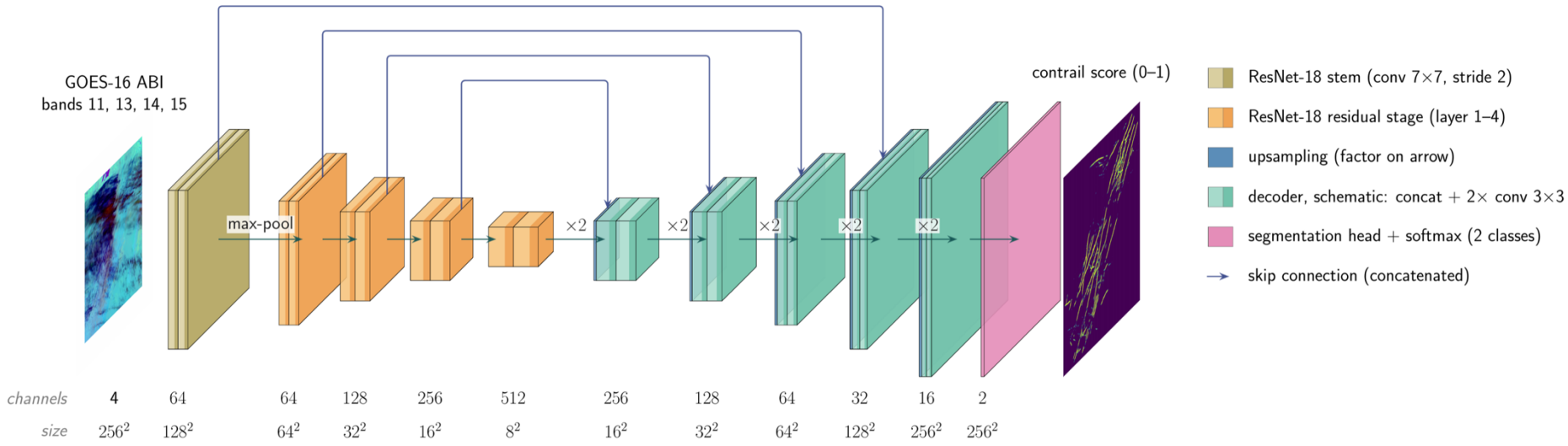
Progress in contrail detection over the past decade

Meijer et al. (2022)

Dice = 0.51

MIT: first large-scale labelled GEO contrail dataset

- **Data:** 103 hand-labelled GOES-16 images over the contiguous US (2000 x 3000 pixels)
- **Algorithm:** U-Net with ResNet-18 encoder



2019–20

Now

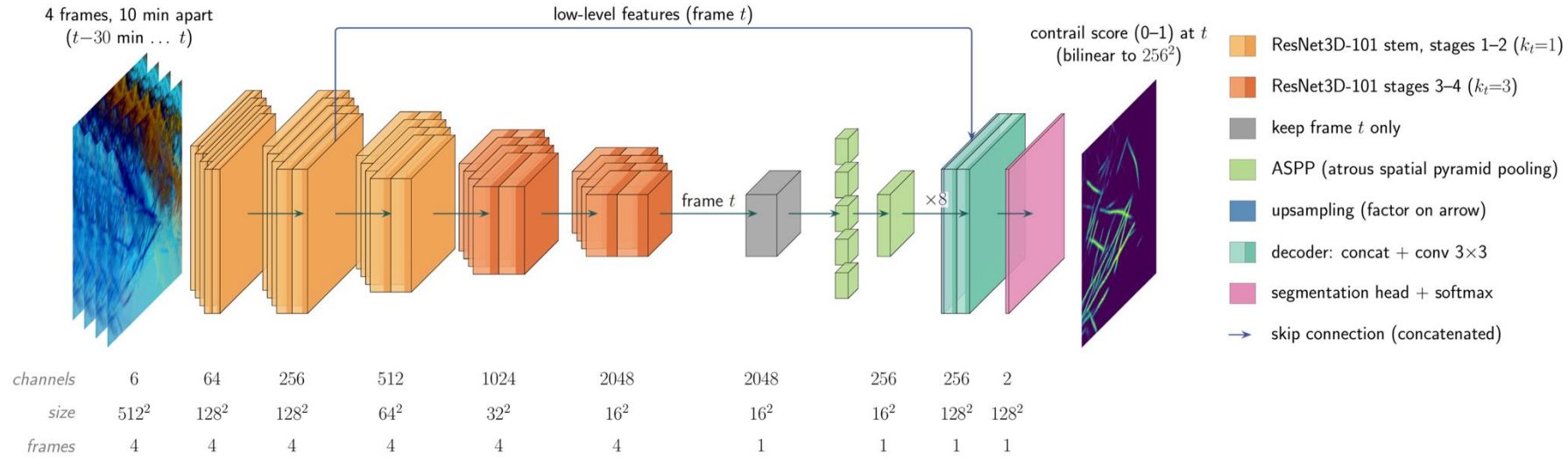
Progress in contrail detection over the past decade

Estimated from Ng et al. (2023)

Dice ≈ 0.66

MIT × Google Research: OpenContrails and improved detection using multi-temporal data

- **Data:** OpenContrails, 22,410 hand-labeled images (256 x 256 pixels)
- **Algorithm:** Use 3 previous images + current with a ResNet3D-101 encoder



2019–20

2020–23

Now

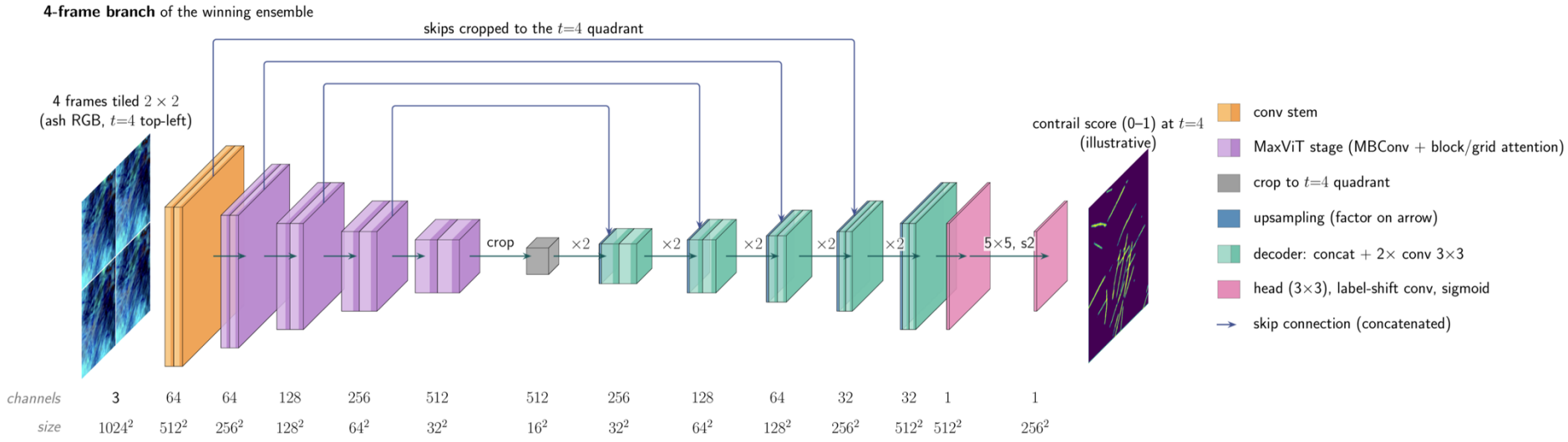
Progress in contrail detection over the past decade

Ng et al. (2023), Kaggle (2023)

Dice = 0.72

OpenContrails + Kaggle competition

- **Data:** OpenContrails, 22,410 hand-labeled images (256 x 256 pixels)
- **Algorithm:** Use 3 previous images + current, but tile them into one larger image. Uses a vision transformer stage.



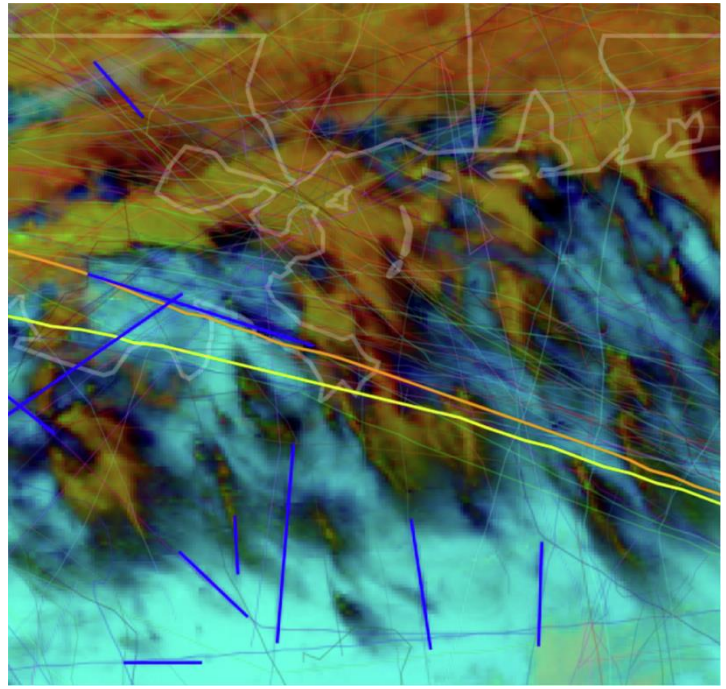
2019–20 2020–23 **2023** Now

Progress in contrail detection over the past decade

Contrail avoidance trials verified using satellite images

- 2023: American Airlines × Google × Breakthrough Energy, 44 flights, 64 % fewer contrails, +2 % fuel on adjusted flights
- 2025, transatlantic follow-up: 62 % fewer contrails on flights that flew the avoidance plan
- Verification: GOES-16 images + automated contrail detection + flight attribution

Sonabend-W et al. (2024); Sankar et al. (2026, preprint)



Flight track (wind-advected) matched to detected contrail



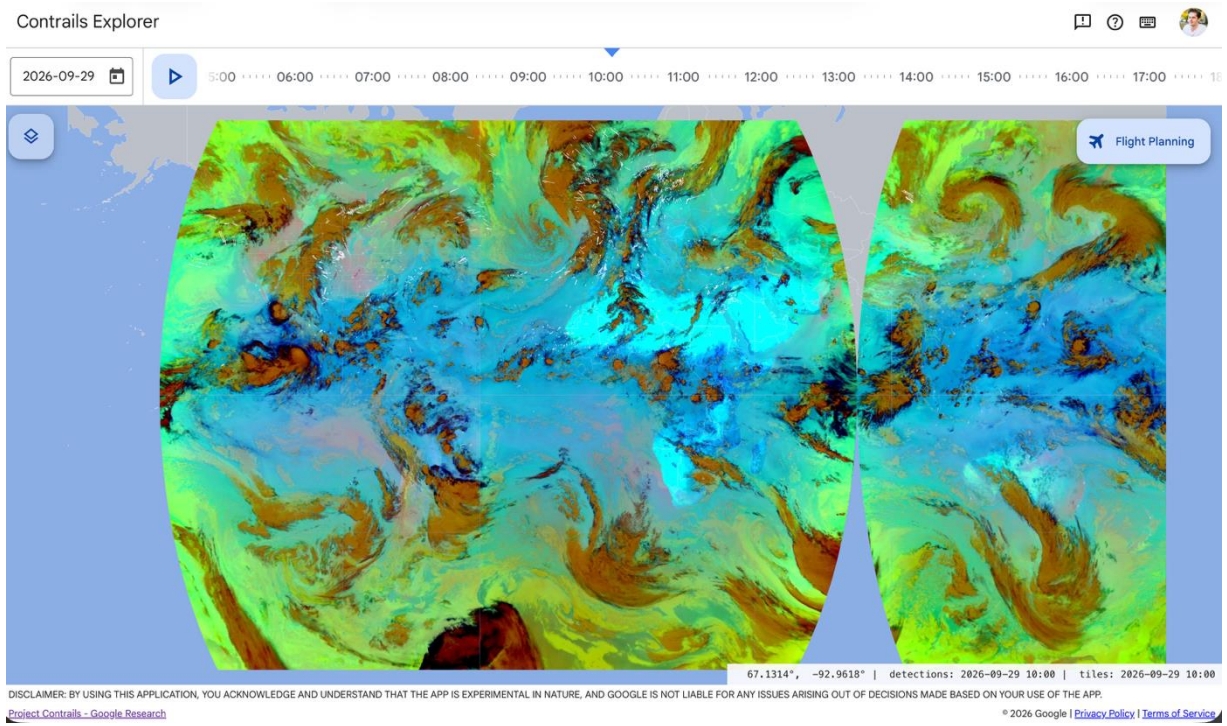
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Progress in contrail detection over the past decade

Google detector goes multi-satellite

- Runs on GOES-East, GOES-West, Himawari and MTG (beta)
- Covers the Americas, Europe/N. Africa and East Asia with 10-min imagery
- Detections served through the Contrails API and the Contrails Explorer web app

contrails.webapps.google.com



Contrails Explorer, 29 Sep 2026: detected contrails (white, very small)

2019–20 | 2020–23 | 2023 | 2023–25 | **2025–26** | Now

That's it! Questions?

If you're interested in this topic and would like to look into it during your MSc thesis and/or a C&O project, reach out to v.r.meijer@tudelft.nl. There are plenty of interesting, open problems that you can work on!

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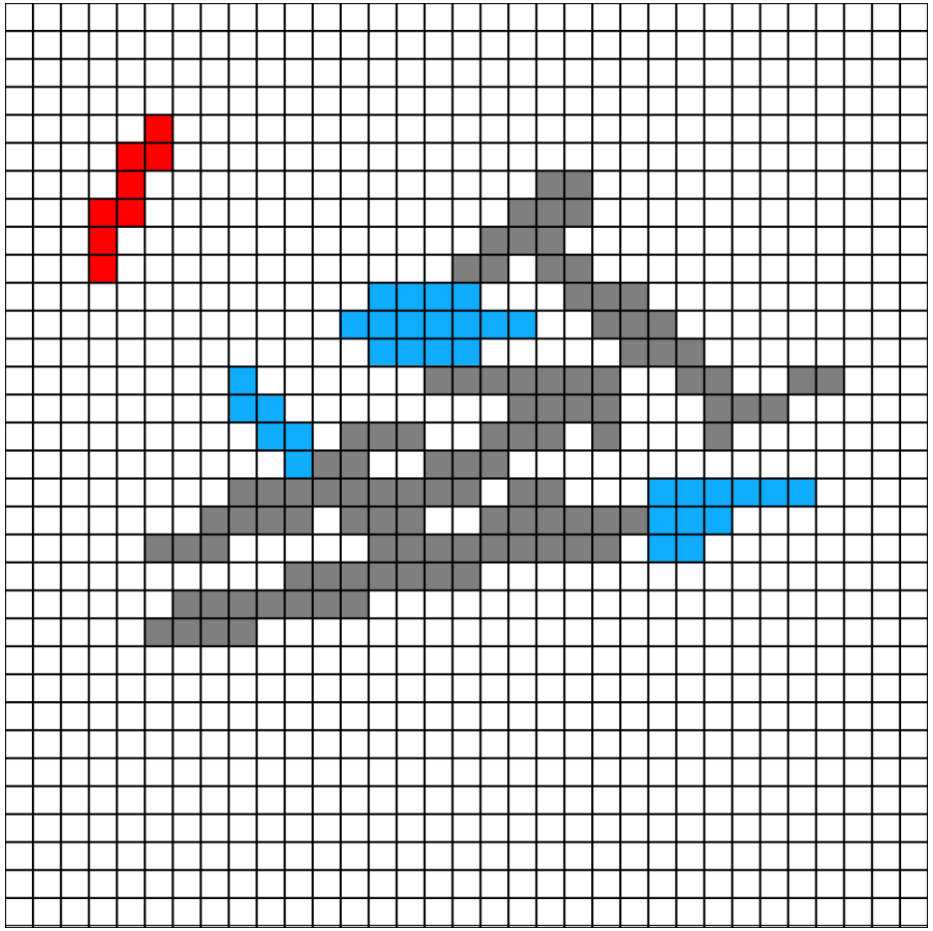
How do we evaluate and characterize the performance of contrail detection algorithms?

Contrail detection: determine which pixels in an image are part of a contrail and which are not.

In computer vision terms, contrail detection is **semantic segmentation** with two classes: contrail and not-contrail.

Detection algorithm errors:

- False positives
- False negatives



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How do we evaluate and characterize the performance of contrail detection algorithms?

Detection algorithm errors:

- False positives (FP)
- False negatives (FN)

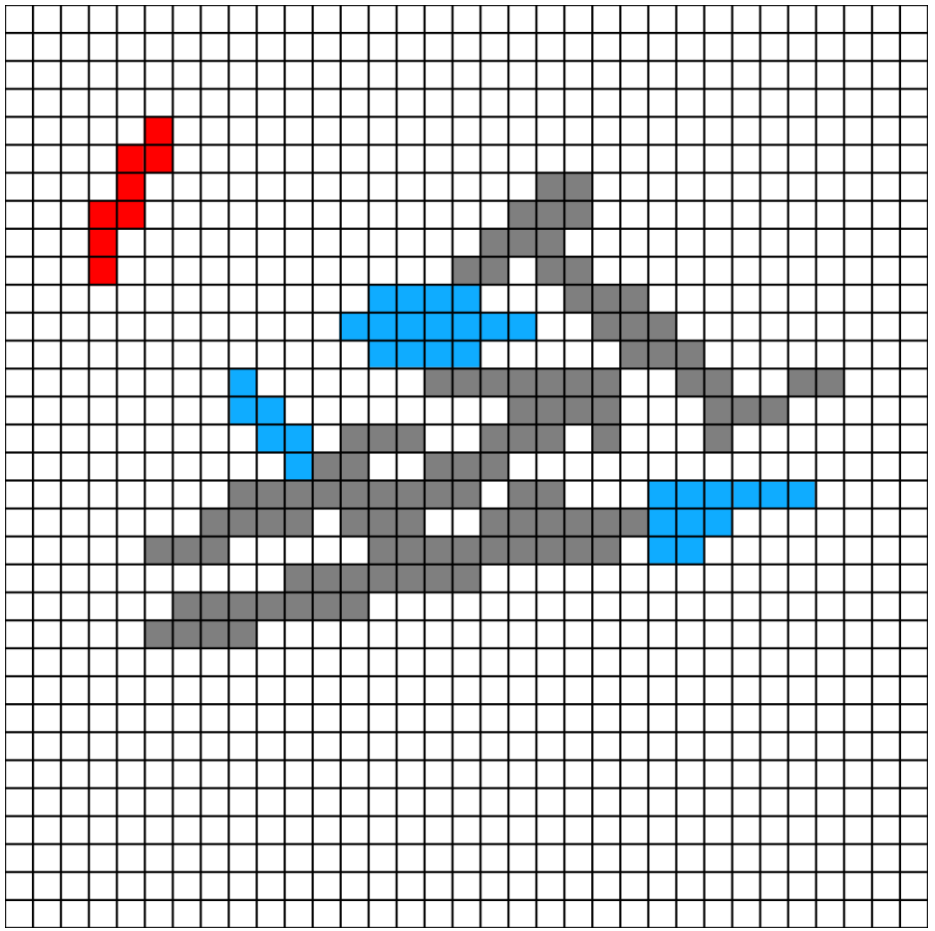
Correct contrail pixels:

- True positives (TP)

Most commonly used metrics:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$



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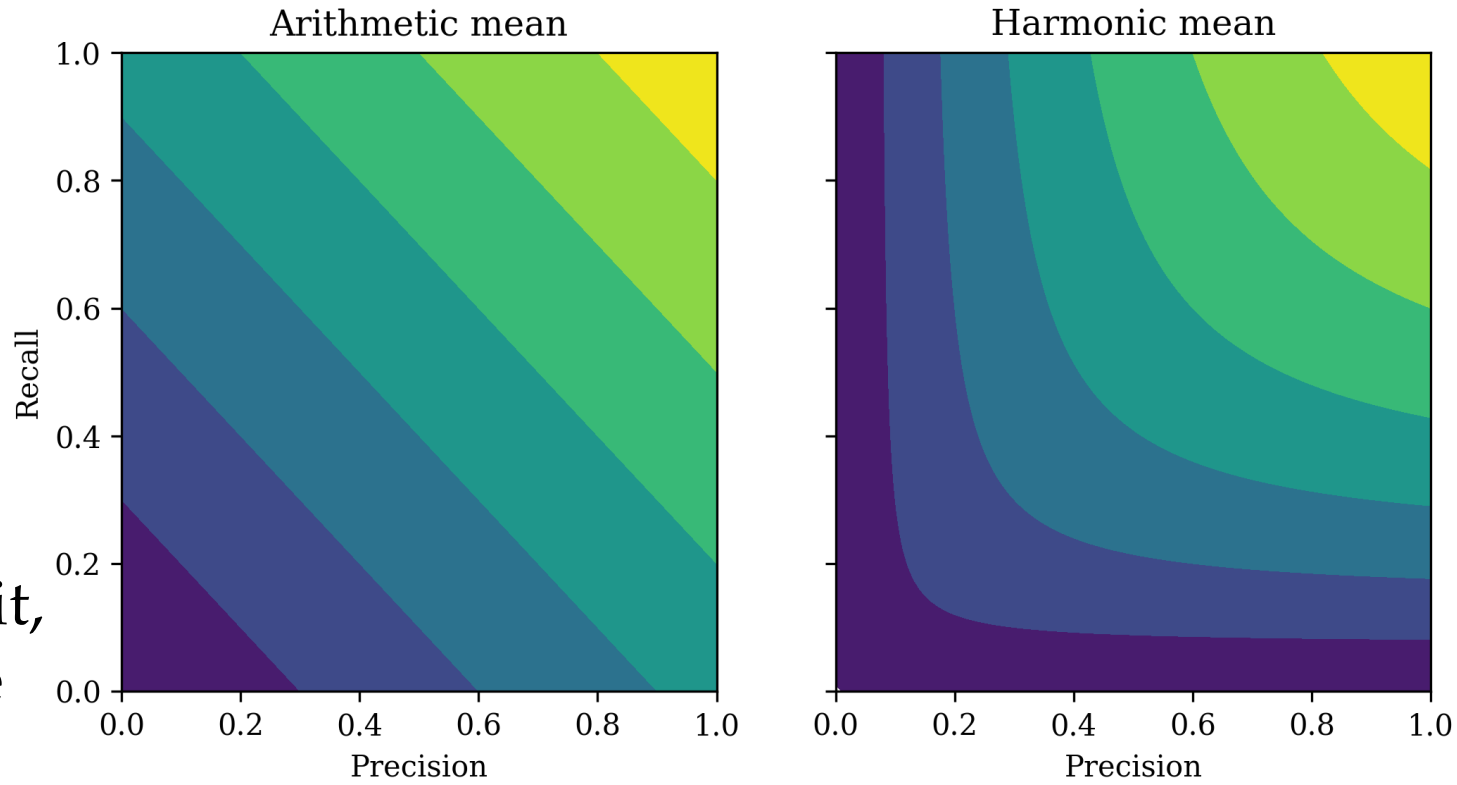
Why use the dice score for evaluation?

Most commonly used metrics:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Precision and recall need to be considered simultaneously.
The **dice score** is one way to do it, as it is the harmonic mean of the precision and recall.



Some tricks of the trade for contrail detection (1/2)

- **Debugging:** when you make any significant changes[1] to your training setup, I advise you to always first run an "overfit experiment", where you try to overfit your model on a very small dataset. Make sure to turn off data augmentation. Your loss values should go to zero very quickly. If not, then there might be an issue with your setup. I've spotted many bugs this way.

- **Visualize:** Make sure to visualize as many relevant metrics and outputs during training as possible. When you're training a semantic segmentation model, you definitely want to visualize both train and validation set examples and the corresponding network outputs over time. This can help you diagnose issues as well.

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1. Significant changes is a bit ill defined. When exploring hyperparameter settings, overfitting may be overkill. But if you're changing architectures or loss functions, you should definitely do this!

Some tricks of the trade for contrail detection (2/2)

- **Class imbalance**: In the grand scheme of things, contrails are "rare", in that they tend to cover less than a 1% of the sky over a domain like the US. This is much lower than natural cloud cover (as you know from living in the Netherlands). So, if you pick any random 256 x 256 pixel tile from a geostationary satellite image, the chances that it doesn't contain any contrails are quite high. This poses issues for data annotation, model training, and evaluation.
  - Make sure that you read up on how the OpenContrails dataset was annotated, to understand decisions that they made to mitigate these issues.
  - For model training, you can mitigate the effects of class imbalance in various ways:
    - you can sample the dataset differently,
    - you can change your network architecture
    - you can modify your loss function.
  - Model evaluation is affected because the precision metric depends on the amount of contrails present in the evaluation dataset: affecting your dice score as well. Therefore, the dice score that you find on your (potentially modified) training set may not carry over directly to the validation and test sets for reasons other than generalization.

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